## Test 2:

### READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

# The Whale Goes to Court

*A remarkable legal drama in the U.S.A*

In 1818, a whale became the subject of a controversial court case in New York City. The case involved an old law requiring those who sold fish oil to pay a fee in order to have their barrels inspected by city officials and certified. However, a fish oil merchant named Samuel Judd refused to pay the inspection fee on three barrels of oil, claiming that no inspection was necessary because it was whale oil, and whales were not fish. The state disagreed, and so a date was set for a court to decide not a matter of law, but the answer to a more fundamental question: Is a whale a fish?

As simple as that question may appear today, the answer was far from obvious in the early nineteenth century. Indeed, the public debate in New York sparked off by the trial was sensational. At stake was nothing less than what most regarded as the order of nature. According to the commonly accepted scheme of things, if an animal was not a beast or a bird, it was a fish, regardless of whether it breathed air or suckled its young. For as long as anyone could remember, all non-human creatures had been organized according to the categories of birds, beasts, or fish. For the average person, the answer seemed perfectly obvious: whales swam in the sea and therefore they were fish.

Against this traditional framework stood the new Linnaean system of classification, which sought to introduce more scientific values into the classification of living things. Barely half a century old in 1818, the Linnaean system controversially classified whales as mammals because they shared two mammalian characteristics: they were warm-blooded and breathed air.

So the trial began in December 1818. Judd's defense lawyers chose as their star witness one of New York's most prominent figures, the congressman Samuel Mitchill. Referred to as a ‘living encyclopedia’ and a ‘walking library’, Mitchill was a renowned natural history lecturer at the College of Physicians and Surgeons who liked to dare his students to test his knowledge of the natural world. “Show me the fin, and I will name the fish,” he boasted. Scientifically, it was an open-and-shut case, and Judd’s defense lawyers believed they only had to do one simple thing to win the trial: insist that it be decided by biology alone. Mitchill seemed fully qualified for this role.

But this is where the case becomes interesting, and is the reason why the trial of the whale is not some dusty nineteenth-century obscurity. For rather than debate Mitchill on his area of

expertise, lawyers for the New York City fish-oil inspectors chose a different tack. Lead counsel William Sampson turned the trial into a contest between scientific learning and common sense, by asking plain-spoken whalers to make their case before the jury.

The crux of Sampson's case was the trial's implication with regard to humans: should Judd be found not guilty? Sampson told jury members that if they accepted Mitchill's testimony on whales, they were obliged to accept a lower place for their own kind in the natural order. As a result, Mitchill's day in court did not go smoothly, not least because he was forced to acknowledge serious disputes among his peers regarding exactly how to classify biological organisms. Sampson then took aim at Mitchill himself and what he represented: privileged aristocrats from the scientific community who had lost touch with reality. The smooth-talking Sampson then claimed Mitchill's beliefs had their origin in Europe – something that he rightly judged would infuriate the citizens of always-independent New York. This revolutionary new thinking, Sampson claimed, was something that honest working New Yorkers could do without. No longer was this a case about barrels of oil; but the proper place of scientific knowledge in the U.S.A.

Ultimately, what won the case for Samuel Judd were not Mitchill's scientific arguments, but testimony from a different sphere altogether. In New York's markets, the merchants implicitly understood that whale oil and fish oil were not the same: whale oil could be used as a fuel for lamps because it could be burned without giving off smoke; fish oil, on the other hand, was nasty, impure stuff used primarily in tanning leather. In the end, it was this testimony that led to Judd's victory.

The trial of the whale still has relevance today, when the court of public opinion remains easily swayed by skeptics. The 1818 trial was really about this question: who gets to decide on the place of the natural world in the human world – scientific experts or public opinion? Whether the issue today is the effects of human activity on the climate of this planet, or one of many other contemporary topics, the legacy of Judd's case continues to be of relevance.

## Questions 1–7

*Do the following statements agree with the information given in Reading Passage 1?*

*In boxes **1–7** on your answer sheet, write:*

| | |
|---|---|
| **TRUE** | if the statement agrees with the information |
| **FALSE** | if the statement contradicts the information |
| **NOT GIVEN** | if there is no information on this |

**1.** An inspection fee on fish oil was introduced in New York in 1818.

**2.** Samuel Judd argued that the inspection fee should exclude whale oil.

**3.** Judd had been in trouble with city officials before the inspection fee disagreement.

**4.** Many New Yorkers were interested in the court case at the time.

**5.** Traditionally, non-human creatures had been classified in one of three groups.

**6.** Generally speaking, ordinary people thought fish were the lowest form of life.

**7.** Whales were excluded from the Linnaean system in 1818.

TRUE IELTS

## Questions 8–13

*Complete the notes below.*

*Choose **ONE WORD ONLY** from the passage for each answer.*

*Write your answers in boxes **8–13** on your answer sheet.*

# The Trial of December 1818

**The Personalities**

- Samuel Mitchill worked as a congressman and **8** _____.
- The defense wanted Mitchill to present the biology of the case.
- William Sampson called **9** _____ as a witness to appeal to the common sense of the jury.

**The Arguments**

- Sampson's case was based on the consequences for humans if Judd won.
- Mitchill admitted that scientists had disputes about classification.
- New Yorkers disliked Mitchill because his ideas came from **10** _____.

**Conclusions**

- In the end, it was statements from local **11** _____, not Mitchill's testimony which helped Judd win.
- Whale oil made a good **12** _____ because it was clean.
- Judd's case is relevant today, e.g., in the debate about Earth's **13** _____.

**ĐÁP ÁN:**

- **1. FALSE** (In 1818 a whale became the subject of a controversial court case in New York City. The case involved an old law requiring those who sold fish oil to pay a fee in order to have their barrels inspected by city officials, and certified. Năm 1818, một con cá voi trở thành chủ đề của một vụ kiện gây tranh cãi tại thành phố New York.Vụ án liên quan đến một đạo luật cũ yêu cầu những người bán dầu cá phải nộp một khoản phí để các thùng dầu của họ được quan chức thành phố kiểm tra và chứng nhận)
- **2. TRUE** (Samuel Judd từ chối trả phí vì cho rằng cá voi không phải là cá).
- **3. NOT GIVEN** (Không có thông tin về việc Judd có gặp rắc rối với chính quyền trước đó hay không).
- **4. TRUE** (Cuộc tranh luận công khai tại New York về phiên tòa được mô tả là "gây chấn động" - *sensational*).
- **5. TRUE** (Theo truyền thống, các sinh vật được phân thành 3 nhóm: chim, thú, hoặc cá).
- **6. NOT GIVEN** (Bài đọc nói người dân nghĩ cá voi là cá, không đề cập đến việc họ coi cá là dạng sống thấp nhất).
- **7. FALSE** (Hệ thống Linnaean thực chất đã phân loại cá voi vào nhóm động vật có vú - *mammals* vào thời điểm đó).
- **8. lecturer** (Referred to as a ‘living encyclopaedia’ and a ‘walking library’, Mitchill was a renowned natural history lecturer at the College of Physicians and Surgeons who liked to dare his students to test his knowledge of the natural world.
- Được gọi là “bách khoa toàn thư sống” và “thư viện biết đi”, Mitchill là một giảng viên lịch sử tự nhiên nổi tiếng tại Trường Cao đẳng Y sĩ và Phẫu thuật, người rất thích thách thức sinh viên kiểm tra kiến thức của ông về thế giới tự nhiên. ).
- **9. whalers** (Lead counsel William Sampson turned the trial into a contest between scientific learning and common sense, by asking plain-spoken whalers to make their case before the jury. Luật sư chính William Sampson đã biến phiên tòa thành một cuộc đối đầu giữa tri thức khoa học và lẽ thường, bằng cách mời những người thợ săn cá voi mộc mạc trình bày quan điểm của họ trước bồi thẩm đoàn. ).

- **10. Europe** (The smooth-talking Sampson then claimed Mitchill's beliefs had their origin in Europe – something he rightly judged would infuriate the citizens of always-independent New York.  Sau đó, Sampson – với tài ăn nói khéo léo – cho rằng những quan điểm của Mitchill có nguồn gốc từ châu Âu, điều mà ông đoán đúng là sẽ khiến những công dân luôn đề cao tính độc lập của New York tức giận. ).
- **11. merchants** (In New York's markets, the merchants implicitly understood that whale oil and fish oil were not the same: whale oil could be used as a fuel for lamps because it could be burned without giving off smoke; fish oil, on the other hand, was nasty, impure stuff used primarily in tanning leather. In the end it was this testimony that led to Judd's victory. Tại các khu chợ ở New York, giới thương nhân ngầm hiểu rằng dầu cá voi và dầu cá không giống nhau: dầu cá voi có thể dùng làm nhiên liệu cho đèn vì khi đốt không tạo ra khói, trong khi dầu cá thì bẩn và không tinh khiết, chủ yếu được dùng trong việc thuộc da. Cuối cùng, chính lời chứng thực này đã dẫn đến chiến thắng của Judd. ).
- **12. fuel** (In New York's markets, the merchants implicitly understood that whale oil and fish oil were not the same: whale oil could be used as a fuel for lamps because it could be burned without giving off smoke; fish oil, on the other hand, was nasty, impure stuff used primarily in tanning leather. In the end it was this testimony that led to Judd's victory. Tại các khu chợ ở New York, giới thương nhân ngầm hiểu rằng dầu cá voi và dầu cá không giống nhau: dầu cá voi có thể dùng làm nhiên liệu cho đèn vì khi đốt không tạo ra khói, trong khi dầu cá thì bẩn và không tinh khiết, chủ yếu được dùng trong việc thuộc da. Cuối cùng, chính lời chứng thực này đã dẫn đến chiến thắng của Judd. ).
- **13. climate** (Whether the issue today is the effects of human activity on the climate of this planet, or one of many other contemporary topics, the legacy of Judd's case continues to be of relevance. Dù vấn đề ngày nay là tác động của hoạt động con người lên khí hậu Trái Đất hay bất kỳ chủ đề đương đại nào khác, di sản của vụ án Judd vẫn tiếp tục có ý nghĩa. ).

## READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14-26**, which are based on Reading Passage 2 below.*

# Why don't we sleep?

*Insomnia, the inability to fall and stay asleep, is at epidemic levels. If sleep is so important to us, why is it sometimes so hard to achieve?*

**A** Fifty-six million prescriptions for sleeping medication were handed out by doctors in America last year. Yet remarkably little is being done to understand insomnia's root causes. Socially and economically, the undertreatment of sleeplessness is hugely expensive. The Institute of Medicine, an independent scientific advisory group, estimates nearly 20 percent of all major accidents on the road are associated with driver sleeplessness. The loss in terms of productivity within the workplace is even higher, costing billions. Then there are the softer costs which remain unmeasured but still exist; for example, the damaged relationships with others. If other medical problems were causing such widespread harm, governments would act, but US authorities contribute only about $230 million a year for the funding of sleep research, comparable to the amount that the manufacturers of popular sleeping pills spent in three months on advertising. Currently most medical school students get no more than four hours of training on sleep disorders; some get none.

**B** If we don't know why we can't sleep, it's in part because we don't really know why we need to sleep in the first place. We know that we divide it between periods of deep-wave sleep and rapid eye movement (REM) sleep, when the brain is as active as when we're awake, but our voluntary muscles are paralyzed. We know that all mammals sleep, and that there are some insects and reptiles that only half sleep, in order to remain aware of their environment and predators.

**C** The predominant theory of sleep is that the brain demands it. Recently, researchers at Harvard, led by Robert Stickgold, tested undergraduates on various aptitude tests, allowed them to nap, then tested them again. They found that those who had engaged in REM sleep subsequently performed better in pattern recognition tasks, such as grammar, while those who slept deeply were better at memorization.

**D** Such studies suggest that memory consolidation may be one function of sleep. Giulio Tononi, a sleep researcher at the University of Wisconsin, published an interesting twist on this theory a few years ago. His study showed that the sleeping brain seems to delete redundant or unnecessary synapses or connections. So the purpose of sleep may be to help us remember what's important, by letting us forget what's not.

**E** Sleep is likely to have physiological purposes too. Some researchers have found that sleep deprivation impedes wound healing in rats, and others have suggested that sleep helps boost the immune system and control infection. But these studies are not conclusive. At Stanford University, William Dement, co-founder of the Stanford Sleep Medicine Center, explains what he knows after 50 years of research, about the reason we sleep. 'As far as I

know,' he said, 'the only reason we need to sleep that is really, really solid is because we get sleepy.'

**F** The Stanford clinic does more than 3,000 overnight sleep studies a year. Their main diagnostic tool is the polysomnogram, the main element of which is the electroencephalograph (EEG), which captures the electrical output from a patient's brain. As the EEG records a person's sleep patterns, the polysomnogram technicians also measure body temperature, muscle activity, eye movement, and breathing. Then they look over the data for signs of abnormal sleep. When a person has narcolepsy, for instance, they plunge from wakefulness into REM sleep without any intermediate steps. However, Clete Kushida, the clinic director, told me he can spot most people's sleep problems right at the intake interview. 'There are those who cannot keep their eyes open, and those who just speak of their exhaustion but don't actually fall asleep. The former often have sleep apnea where they stop breathing. The latter have true insomnia.'

**G** In sleep apnea, muscle relaxation allows the soft tissue of the throat to close, shutting off the sleeper's air passage. When the brain realizes it is not getting oxygen, it sends a signal to the body to wake up. The sleeper awakes, inhales, and sleep returns. Apnea is a serious problem but only indirectly a sleep disease. True insomniacs are people who either can't get to, or can't stay asleep for any evident reason.

**H** While apnea can be treated with a device that forces air down the sleeper's throat to keep the airways open, the treatment of classic insomnia is not so clear cut. Acupuncture may help. It has long had this role in Asian medicine and is being studied at the University of Pittsburgh sleep center now. Typically in the US, insomnia is treated in two ways. First come the sleeping pills. Though safer than they once were, they can still lead to psychological addiction. 'Many users complain that their sleeping-pill sleep seems different, and they feel unfocused on waking. Sleeping pills are not a natural way to sleep,' points out Charles Czeisler, director of the Harvard Health and Safety Group.

**I** The second step in treating insomniacs is usually cognitive behavioral therapy (CBT). In CBT a psychologist teaches insomniacs to think about sleep problems as manageable, and to practice good 'sleep hygiene': sleep in a dark room, go to bed only when you are sleepy, don't exercise beforehand. Studies have shown that CBT is more effective than sleeping pills at treating long-term insomnia, but many sufferers aren't convinced. 'Some people continue in my experience to struggle,' says John Winkelman, another sleep center director. 'They're not super-satisfied with their sleep.'

## Questions 14 – 17

*Complete the summary below.*

*Choose **ONE WORD ONLY** from the passage for each answer.*

*Write your answer in boxes 14-17 on your answer sheet.*

### Insomnia in the USA

Last year the high number of **14** ..... doctors gave their patients highlights the fact that insomnia is a serious issue in the USA. It is evidently responsible for a number of serious traffic **15** ..... and has a considerable effect on company **16** ..... . A lack of sleep can also have a harmful effect on different relationships, although there are no statistics on this. Unfortunately, there is insufficient official funding given to research into sleep problems, and the **17** ..... given at medical school is also insufficient to deal with insomnia.

## Questions 18 – 23

Reading Passage 2 has nine paragraphs, **A-I**.

Which paragraph contains the following information?

*Write the correct letter, **A-I**, in boxes 18-23 on your answer sheet.*

**18** a mention of different cultural approaches to dealing with insomnia

**19** indications that sleep has a positive effect on health

**20** the effects of different kinds of sleep on mental performance

**21** reasons why various species of animal sleep in different ways

**22** the suggestion that sleep allows us to get rid of insignificant information

**23** a description of how the brain responds when a person stops breathing

## Questions 24 – 26

*Look at the following statements (Questions 24-26) and the list of people below.*

*Match each statement with the correct person,* ***A-E****.*

*Write the correct letter,* ***A-E****, in boxes 24-26 on your answer sheet.*

**24** Research tells us that we sleep as a reaction to our body's signals.

**25** People suffering from insomnia are easily recognizable

**26** Therapy does not always enable people to achieve good quality sleep.

## List of Researchers

**A** Giulio Tononi
**B** William Dement
**C** Clete Kushida
**D** Charles Czeisler
**E** John Winkelman

TRUE IELTS

## ĐÁP ÁN:

**14. prescriptions** (Số lượng lớn đơn thuốc mà bác sĩ kê cho bệnh nhân phản ánh mức độ nghiêm trọng của chứng mất ngủ).

***Đoạn chứa đáp án***:

Fifty-six million prescriptions for sleeping medication were handed out by doctors in America last year.

**15. accidents** (Mất ngủ là nguyên nhân gây ra nhiều vụ tai nạn giao thông nghiêm trọng).

***Đoạn chứa đáp án***:

The Institute of Medicine, an independent scientific advisory group, estimates nearly 20 percent of all major accidents on the road are associated with driver sleepiness.

**16. productivity** (Tình trạng này ảnh hưởng đáng kể đến năng suất của các công ty).

***Đoạn chứa đáp án***:

The loss in terms of productivity within the workplace is even higher, costing billions.

**17. training** (Quá trình đào tạo tại các trường y cũng không đủ để giải quyết vấn đề này).

***Đoạn chứa đáp án***:

Currently most medical school students get no more than four hours of training on sleep disorders; some get none.

**18. H** (Đề cập đến các phương pháp văn hóa khác nhau như châm cứu trong y học châu Á).

***Đoạn chứa đáp án***:

Passage H: While apnea can be treated with a device that forces air down the sleeper's throat to keep the airways open, the treatment of classic insomnia is not so clear cut. Acupuncture may help. It has long had this role in Asian medicine and is being studied at the University of Pittsburgh sleep center now.

**19. E** (Chỉ ra giấc ngủ có tác dụng tích cực như làm lành vết thương và tăng cường hệ miễn dịch).

***Đoạn chứa đáp án***:

Passage E: Sleep is likely to have physiological purposes too. Some researchers have found that sleep deprivation impedes wound healing in rats, and others have suggested that sleep helps boost the immune system and control infection.

**20. C** (Tác động của các loại giấc ngủ khác nhau như REM hay ngủ sâu đối với các bài kiểm tra năng lực trí tuệ).

***Đoạn chứa đáp án***:

Passage C: They found that those who had engaged in REM sleep subsequently performed better in pattern recognition tasks, such as grammar, while those who slept deeply were better at memorization.

**21. B** (Lý do các loài động vật khác nhau ngủ theo những cách khác nhau, ví dụ để cảnh giác kẻ thù).

***Đoạn chứa đáp án***:

Passage B: We know that all mammals sleep, and that there are some insects and reptiles that only half sleep, in order to remain aware of their environment and predators.

**22. D** (Gợi ý rằng giấc ngủ cho phép bộ não xóa bỏ các kết nối thần kinh dư thừa/không quan trọng).

***Đoạn chứa đáp án***:

Passage D: His study showed that the sleeping brain seems to delete redundant or unnecessary synapses or connections. So the purpose of sleep may be to help us remember what's important, by letting us forget what's not.

**23. G** (Mô tả cách não phản ứng khi một người ngừng thở trong chứng ngưng thở khi ngủ).

***Đoạn chứa đáp án***:

Passage G: In sleep apnea, muscle relaxation allows the soft tissue of the throat to close, shutting off the sleeper's air passage. When the brain realizes it is not getting oxygen, it sends a signal to the body to wake up. The sleeper awakes, inhales, and sleep returns.

**24. B** (William Dement: "Lý do duy nhất chúng ta cần ngủ... là vì chúng ta thấy buồn ngủ").

***Đoạn chứa đáp án***:

Passage E: 'As far as I know,' he said, 'the only reason we need to sleep that is really, really solid is because we get sleepy.'

**25. C** (Clete Kushida: Có thể nhận diện hầu hết vấn đề về giấc ngủ ngay tại cuộc phỏng vấn đầu vào qua đôi mắt hoặc cách nói chuyện).

***Đoạn chứa đáp án***:

Passage F: However, Clete Kushida, the clinic director, told me he can spot most people's sleep problems right at the intake interview. 'There are those who cannot keep their eyes open, and those who just speak of their exhaustion but don't actually fall asleep. The former often have sleep apnea where they stop breathing. The latter have true insomnia.'

**26. E** (John Winkelman: Liệu pháp tâm lý không phải lúc nào cũng giúp mọi người hài lòng với giấc ngủ của mình).

***Đoạn chứa đáp án***:

Passage I: 'Some people continue in my experience to struggle,' says John Winkelman, another sleep center director. 'They' re not super-satisfied with their sleep.'

# READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

## The Voynich Manuscript

The starkly modern Beinecke Library at Yale University is home to some of the most valuable books in the world: first folios of Shakespeare, Gutenberg Bibles and manuscripts from the early Middle Ages. Yet the library's most controversial possession is an unprepossessing vellum manuscript about the size of a hardback book, containing 240-odd pages of drawings and text of unknown age and authorship. Catalogued as MS 408, the manuscript would attract little attention were it not for the fact that the drawings hint at esoteric knowledge, while the text seems to be some sort of code — one that no-one has been able to break. It is known to scholars as the Voynich manuscript, after the American book-dealer Wilfrid Voynich, who bought the manuscript from a Jesuit college in Italy in 1912.

Over the years, the manuscript has attracted the attention of everyone from amateur dabblers to top code-breakers, all determined to succeed where countless others have failed. Academic research papers, books and websites are devoted to making sense of the contents of the manuscript, which are freely available to all. "Most other mysteries involve second-hand reports," says Dr Gordon Rugg of Keele University, a leading Voynich expert. "But this is one that you can see for yourself."

It is certainly strange: page after page of drawings of weird plants, astrological symbolism and human figures, accompanied by a script that looks like some form of shorthand. What does it say — and what are the drawings about? Voynich himself believed that the manuscript was the work of the 13th-century English monk Roger Bacon, famed for his knowledge of alchemy, philosophy and science. In 1921 Voynich's view that Bacon was the writer appeared to win support from the work of William Newbold, Professor of Philosophy at the University of Pennsylvania, who claimed to have found the key to the cipher system used by Bacon. According to Newbold, the manuscript proved that Bacon had access to a microscope centuries before they were supposedly first invented. The claim that this medieval monk had observed living cells created a sensation. It soon became clear, however, that Newbold had fallen victim to wishful thinking. Other scholars showed that his "decoding" methods produced a host of possible interpretations.

The Voynich manuscript has continued to defy the efforts of world-class experts. In 1944, a team was assembled to tackle the mystery, led by William Friedman, the renowned American code-breaker. They began with the most basic code-breaking task: analysing the relative frequencies of the characters making up the text, looking for signs of an underlying structure. Yet Friedman's team soon found themselves in deep water. The precise size of the "alphabet" of the Voynich manuscript was unclear: it is possible to make out more than 70 distinct symbols among the 170,000-character text. Furthermore, Friedman discovered that some words and phrases appeared more often than expected in a standard language, casting doubt on claims that the manuscript concealed a real language, as encryption typically reduces word frequencies.

Friedman concluded that the most plausible resolution of this paradox was that “Voynichese” is some sort of specially created artificial language, whose words are devised from concepts rather than linguistics. So could the Voynich manuscript be the earliest-known example of an artificial language? “Friedman’s hypothesis commands respect because of the lifetime of cryptanalytic expertise he brought to bear,” says Rob Churchill, co-author of *The Voynich Manuscript*. That still leaves a host of questions unanswered, however, such as the identity of the author and the meaning of the bizarre drawings. “It does little to advance our understanding of the manuscript as a whole,” says Churchill.

Even though Friedman was working more than 60 years ago, he suspected that major insights would come from using the device that had already transformed code-breaking: the computer. In this he was right — it is now the key tool for uncovering clues about the manuscript’s language.

The insights so far have been perplexing. For example, in 2001 another leading Voynich scholar, Dr Gabriel Landini of Birmingham University in the UK, published the results of his study of the manuscript using a pattern-detecting method called spectral analysis. This revealed evidence that the manuscript contains genuine words, rather than random nonsense, consistent with the existence of some underlying natural language. Yet the following year, Voynich expert René Zandbergen of the European Space Agency in Darmstadt, Germany, showed that the entropy of the text (a measure of the rate of transfer of information) was consistent with Friedman’s suspicions that an artificial language had been used.

Many are convinced that the Voynich manuscript is not a hoax. For how could a medieval hoaxer create so many tell-tale signs of a message from random nonsense? Yet even this has been challenged in new research by Rugg. Using a system first published by the Italian mathematician Girolamo Cardano in 1150, in which a specially constructed grille is used to pick out symbols from a table, Rugg found he could rapidly generate text with many of the basic traits of the Voynich manuscript. Publishing his results in 2004, Rugg stresses that he had not set out to prove the manuscript a hoax. “I simply demonstrated that it is feasible to hoax something this complex in a few months,” he says.

Inevitably, others beg to differ. Some scholars, such as Zandbergen, still suspect the text has genuine meaning, though believe it may never be decipherable. Others, such as Churchill, have suggested that the sheer weirdness of the illustrations and text hints at an author who had lost touch with reality.

What is clear is that the book-sized manuscript, kept under lock and key at Yale University, has lost none of its fascination. “Many derive great intellectual pleasure from solving puzzles,” says Rugg. “The Voynich manuscript is as challenging a puzzle as anyone could ask for.”

## Questions 27–30

*Do the following statements agree with the information given in Reading Passage 3?*

*In boxes 27–30 on your answer sheet, write*

**TRUE** — if the statement agrees with the information
**FALSE** — if the statement contradicts the information
**NOT GIVEN** — if there is no information on this

**27** It is uncertain when the Voynich manuscript was written.

**28** Wilfrid Voynich donated the manuscript to the Beinecke Library.

**29** Interest in the Voynich manuscript extends beyond that of academics and professional code-breakers.

**30** The text of the Voynich manuscript contains just under 70 symbols.

## Questions 31–34

*Look at the following statements (Questions 31–34) and the list of people below.*

*Match each statement with the correct person,* ***A–H****.*

*Write the correct letter,* ***A–H****, in boxes 31–34 on your answer sheet.*

**31** The number of times that some words occur makes it unlikely that the manuscript is based on an authentic language.

**32** Unlike some other similar objects of fascination, people can gain direct access to the Voynich manuscript.

**33** The person who wrote the manuscript may not have been entirely sane.

**34** It is likely that the author of the manuscript is the same person as suggested by Wilfrid Voynich.

### List of People

**A** Gordon Rugg
**B** Roger Bacon
**C** William Newbold
**D** William Friedman
**E** Rob Churchill
**F** Gabriel Landini
**G** René Zandbergen
**H** Girolamo Cardano

## Questions 35–39

*Complete the summary below.*

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

*Write your answers in boxes 35–39 on your answer sheet.*

### Voynich Researchers

William Newbold believed that the author of the Voynich manuscript had been able to look at cells through a **35** _____. Other researchers later demonstrated that there were flaws in his argument.

William Friedman concluded that the manuscript was written in an artificial language that was based on **36** _____. He couldn't find out the meaning of this language but he believed that the **37** _____ would continue to bring advances in code-breaking.

Dr Gabriel Landini used a system known as **38** _____ in his research, and claims to have demonstrated the presence of genuine words.

Dr Gordon Rugg's system involved a grille, that made it possible to quickly select symbols that appeared in a **39** _____. Rugg's conclusion was that the manuscript lacked genuine meaning.

## Question 40

*Choose the correct letter, **A, B, C or D**.*

*Write the correct letter in box 40 on your answer sheet.*

The writer's main aim in this passage is to

**A** explain the meaning of the manuscript.
**B** determine the true identity of the manuscript's author.
**C** describe the numerous attempts to decode the manuscript.
**D** identify which research into the manuscript has had the most media coverage.

## ĐÁP ÁN:

**27. TRUE**

*Giải thích:* Đoạn 1 nêu rõ bản thảo có "độ tuổi và tác giả chưa được biết đến" (unknown age and authorship).
"...containing 240-odd pages of drawings and text of unknown age and authorship."

**28. NOT GIVEN**

*Giải thích:* Văn bản chỉ nói Wilfrid Voynich đã mua bản thảo vào năm 1912, không đề cập đến việc ông có tặng nó cho Thư viện Beinecke hay không
Trích dẫn: *"The manuscript surfaced in 1912 when it came into the hands of Wilfrid Voynich... It now resides in Yale University's Beinecke Library."* (Đoạn 1) .

**29. TRUE**

*Giải thích:* Đoạn 2 cho biết bản thảo thu hút sự chú ý của tất cả mọi người, từ "những người nghiệp dư" (amateur dabblers) đến các chuyên gia giải mã hàng đầu.
Trích dẫn: *"...has attracted the attention of everyone from amateur dabblers to top codebreakers."* (Đoạn 2)

**30. FALSE**

*Giải thích:* Đoạn 4 đề cập rằng có thể nhận ra "hơn 70 ký tự riêng biệt" (more than 70 distinct symbols), không phải "ít hơn 70".
Trích dẫn: *"The script uses more than 70 distinct symbols..."* (Đoạn 4)

**31. D (William Friedman):** Ông phát hiện một số từ xuất hiện thường xuyên hơn dự kiến, gây nghi ngờ về việc đây có phải là một ngôn ngữ thật hay không.
Trích dẫn: *"...some words and phrases appeared more often than expected..."* (Đoạn 4)

**32. A (Gordon Rugg):** Ông nói rằng trong khi các bí ẩn khác thường qua lời kể gián tiếp, bí ẩn này là thứ "bạn có thể tự mình nhìn thấy" (freely available).
Trích dẫn: *"...this is one that you can see for yourself."* (Đoạn 2)

**33. E (Rob Churchill):** Ông gợi ý rằng sự kỳ lạ của hình minh họa và văn bản có thể cho thấy tác giả là người "mất liên lạc với thực tại" (lost touch with reality).
Trích dẫn: *"...hint at an author who had lost touch with reality."* (Đoạn 9)

**34. C (William Newbold):** Ông ủng hộ quan điểm của Voynich rằng Roger Bacon chính là tác giả của bản thảo.
Trích dẫn: *"...appeared to win support from several quarters."* (Đoạn 3)

**35. microscope:** Newbold cho rằng bản thảo chứng minh Bacon đã tiếp cận với kính hiển vi trước khi nó được phát minh.
Trích dẫn: *"According to Newbold, the manuscript proved that Bacon had access to a microscope centuries before they were supposedly first invented."*

**36. concepts:** Friedman kết luận đây là một ngôn ngữ nhân tạo được tạo ra từ các khái niệm (concepts) thay vì ngôn ngữ học.
Trích dẫn: *"...whose words are devised from concepts, rather than linguistics."*

**37. computer:** Friedman tin rằng máy tính sẽ tiếp tục mang lại những tiến bộ trong việc giải mã.
Trích dẫn: *"...he suspected that major insights would come from using the device that had already transformed codebreaking: the computer."*

**38. spectral analysis:** Dr Gabriel Landini đã sử dụng phương pháp phân tích quang phổ trong nghiên cứu của mình.
Trích dẫn: *"...published the results of his study of the manuscript using a pattern-detecting method called spectral analysis."*

**39. table:** Hệ thống của Gordon Rugg sử dụng một chiếc vỉ (grille) để chọn nhanh các biểu tượng xuất hiện trong một bảng (table).
Trích dẫn: *"...in which a specially constructed grille is used to pick out symbols from a table."*

**40. C (describe the numerous attempts to decode the manuscript)**

*Giải thích:* Toàn bộ bài đọc tập trung mô tả các nỗ lực khác nhau của nhiều chuyên gia và tổ chức trong việc cố gắng giải mã bản thảo Voynich qua nhiều thập kỷ.
Lý do: Cả bài là một tổng quan về nhiều nhà nghiên cứu khác nhau, các phương pháp và giả thuyết của họ về bản thảo Voynich.

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